

Blended Collaboration: Communication and Cooperation Between Two Users Across the Reality-Virtuality Continuum

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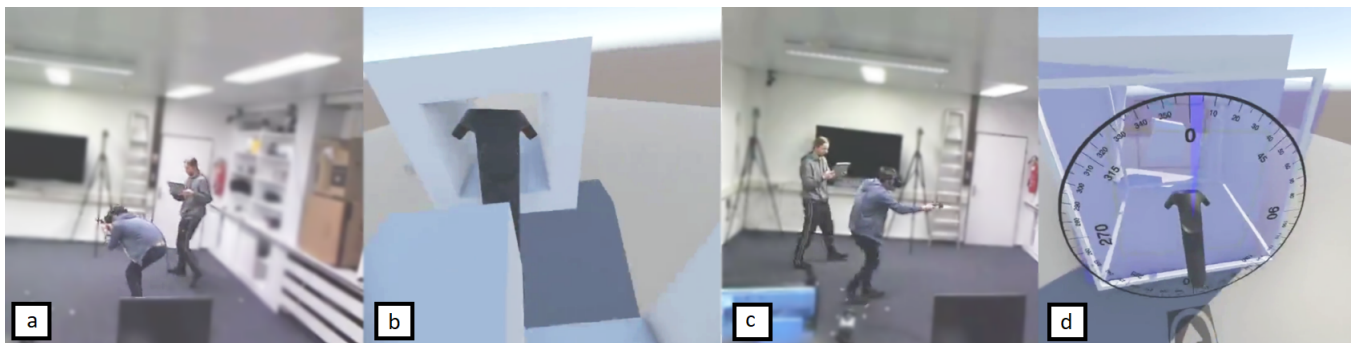


Figure 1: Asymmetric interaction to maneuver a virtual object through a tunnel while avoiding collisions; the VR user can only see the obstacles and specifies translations (a – c), the AR user can only scale and rotate the object and is guided by the VR user (d). Both users must, therefore, complete the task collaboratively.

ABSTRACT

Mixed reality (MR) technologies provide enormous potential for collaboration between multiple users across the Reality–Virtuality continuum. We evaluate communication in a MR-based two-user collaboration task, in which the users have to move an object through an obstacle without collision. We used a blended reality environment, in which one user is immersed in virtual reality, whereas the other uses mobile augmented reality. Both users have different abilities and information and mutually depend on each other for successful completion of the task. Communication consensus can

either be achieved by using speech, visual widgets, or a combination of both. The results indicate that speech plays a fundamental role. The usage of widgets served as an extension rather than a replacement of language. However, the combination of speech and widgets improved the clearness of communication with less miscommunication. These results provide important indications about how to design blended collaboration across the Reality–Virtuality continuum.

CCS CONCEPTS

• **Human-centered computing** → **User studies; Mixed / augmented reality; Virtual reality.**

KEYWORDS

virtual reality, augmented reality, cooperation, collaboration, 3D interaction, communication, multi-user XR

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1 INTRODUCTION

The Reality–Virtuality continuum is a continuous scale ranging between fully computer-generated virtual environments (VEs), the *virtuality*, and the completely physical *reality* [16]. Within the continuum, all possible mixed reality (MR) combinations of real and virtual objects are possible. Recently, MR technologies such as augmented reality (AR) and virtual reality (VR) have advanced enormously, and are now widely available in the consumer market. Today, VR applications range from entertainment to professional practices. AR blends physical reality and virtual information by superimposing the user's field of view in the real world with virtual objects. The user can seamlessly manipulate virtual objects within the real world, and at the same time use the real world as part of the user interface. Such AR user interfaces can amplify human perception and cognition of the real world [29]. In contrast, VR environments allow immersing a user fully into a computer-generated environment. VR technology typically enables a better spatial understanding and higher sense of presence compared to other media [30]. In the scope of the reality-virtuality continuum, many works focused on user cooperation following the sum-of-contributions approach for collaborative manipulations [6]. The cooperation can happen within the same spectrum of the Reality–Virtuality continuum, like VR-VR [18, 23] or AR-AR [19, 21]. Hardware add-ons can also facilitate the inclusion of AR users in the vicinity while enabling asymmetric interaction with VR counterparts [11]. While head-mounted displays (HMDs) isolate the user from the real world around them [28], integrating them into a collaborative virtual environment (CVE) can lead to a unique collaboration experience, in particular, when combined with a video pass-through for MR interactions [5, 19, 21, 24–27, 31]. Nevertheless, each technology comes with its own limits – restricted perception of the real world for VR users and restricted representation of the user within the virtual world for AR users.

When two or more users work together to solve a task, it is particularly important for them to be able to communicate with each other, to find common ground and maneuver around the limitations of each technology. Co-located AR collaborators benefit from a high degree of spatial awareness because of the shared physical environment, and shared reference points [17, 18]. However, in case different devices are used such as a VR and AR, users might misperceive what the other user is seeing [21]. In contrast, a lot of the above-mentioned spatial cues are missing in VR. Here, a virtual body of the user can improve spatial awareness by providing cues such as body and hand gestures, 3D pointers or to indicate the other user's gaze or focus [12, 22, 23, 25, 32, 33], but due to limited tracking, these cues might be missing for the AR user – especially when using technology such as tablets or smartphones. It has been

shown that such body cues can facilitate communication and result in faster completion times of cooperative tasks [33].

In some cases, one could imagine that the virtual body of a bystander would even hinder the experience for the VR user. For example, when working with older adults that play a VR game, assistants are often present to help them if they have any questions or to control the game experience [13]. For them, it is important to perceive what the VR user is seeing, but for the VR user, the presence of another person might not always make sense within the scope of the game. In scenarios like these, where there is no virtual body for the AR user, other visual communication cues like notes, sketches, or annotations can be used [8, 9, 12, 19], but it should be easy to map guiding actions to a (real) 3D task [14]. When looking at natural, verbal communication, it can be seen that users often talk to each other and verbally plan their work in a collaborative task [8, 24, 33], but that effective spatial referencing can be challenging without the use of visual cues [19]. Nonverbal aspects of communication play a vital role in natural human disclosure, and they highlight the differences in gesturing strategies either with, or without using speech [33].

While there are studies that combine both technologies in a single setup [5, 7, 19, 21, 24, 26, 31], the importance of verbal communication and widgets within a collaborative task between AR and VR users has to be studied in more depth.

The contributions of this paper can be summarized as follows:

- (1) A detailed analysis of the use of verbal communication for solving a cooperative VR/AR task,
- (2) the development of visual widgets to facilitate the communication about spatial information, and
- (3) a comparison of the speech only, visual widgets only and the combination during a cooperative task.

2 STUDY

The main task during the study was to move an object through an obstacle, without the object touching it. During this task, the participants have to work together, since one person can only translate the object's position and the other one is responsible for scaling and rotating the object (see Figure 1 a – c). Communication can either be established through natural speech or through a set of different widgets that can be placed by one participant and seen by the other one (see Figure 1 d). While not using verbal communication is a rare scenario, it is possible in the context of users speaking different languages, in quiet environments or when there is a lot of background noise making verbal communication difficult. For example, users in gas pressure regulation stations have to face loud ambient noise, which can interfere with verbal commands [10].

The study consists of three different conditions for solving an asymmetric collaboration task. All conditions were tested in a within-subject design and were presented in a pseudo-randomized order using a Latin square.

- **Speech-Only:** In this condition, the participants can only use natural speech for communication. No widgets can be placed and seen.

- **Widgets-Only:** In this condition, the participants are not allowed to use natural speech to communicate with each other. The only indication that the AR participant gets about what to do, are the widgets provided by the VR participant.
- **Speech-and-Widgets:** This condition is a combination of conditions 1 and 2. The participants can use both natural speech and widgets to communicate with each other.

The goal of this study is to evaluate the effectiveness and the change of communication patterns when using visual widgets, in a collaborative XR environment.

Our hypotheses are:

- H1: Conditions using natural speech will be faster,
- H2: Conditions using widgets will be more precise,
- H3: The combination of speech and widgets will result in the highest usability, and
- H4: Widgets will result in less miscommunication compared to verbal communication.

For the experiment, we used an HTC Vive Pro with one controller for the participant in VR, on a computer with a GeForce GTX 1080 Ti graphics card. The AR participant used an iPad Pro. While VR HMDs have gotten more common for private use in the last few years, AR HMDs are still mostly used in industry or research. Therefore, we chose this tablet-based, video AR instead of an HMD-based, see-through AR setup. Nevertheless, the tablet-based AR comes with limited tracking of the AR user and the need to hold the tablet in their hands, preventing them from using real-life gestures. Furthermore, limited depth cues can prevent the user from perceiving the virtual image correctly. These are some of the reasons why we supposed that widgets might be useful in this scenario, rather than using different technology.

The application was developed with the Unity game engine version 2018.2.13, and the networking between the two devices was implemented with Photon SDK [4]. Verbal communication between the two users was possible since both were in the same room during the experiment.

2.1 Application Setup

Every trial contained a wall (i.e., obstacle) and the *movable object*. The users' task was to guide the *movable object* through the wall without collision, followed by a next trial. Participants completed a set of nine different trials, with every trial containing a wall with one rotation around the x-, y-, or z-axis and a scale difference. The walls as they appear to the VR user are shown in Figure 2 (a). While the VR participant always sees the whole wall with the hole for the *movable object*, this hole is hidden for the AR participant, who only sees the outline of the wall. We chose to restrict the AR user's knowledge to mimic an expert-student scenario and to increase communication between the two users, where the VR user serves as the expert and the AR user as the student. In case of a collision with the obstacle, the part of the wall that was touched turns red for two seconds and the cube's position is reset, while its orientation and scale as set by the users remain the same. We chose this technique to avoid users translating the *movable object* through the wall without paying attention to the number of collisions they made.

Since we want to examine the collaborative work between users in VR and AR, the task during the study needs to be asymmetric.

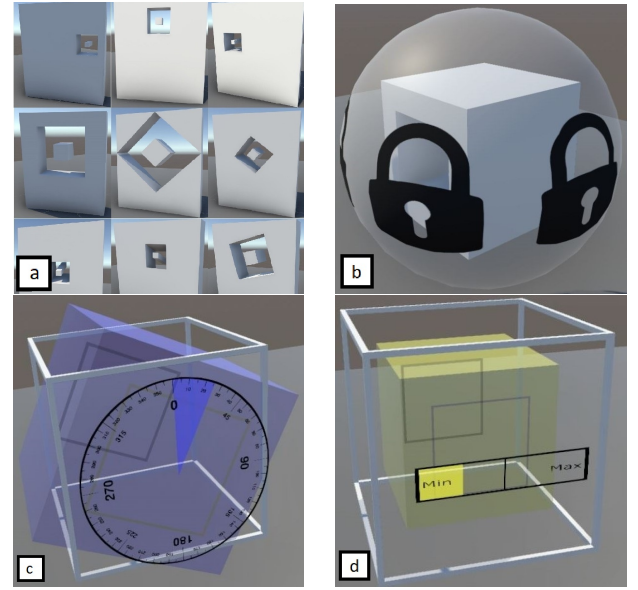


Figure 2: From upper left to lower right: All obstacles used in the study; some are rotated around the y-axis, hence the different perspectives (a), the locking sphere used to prevent concurrent user input (b), the VR widget view for rotation (c) and scaling (d).

We are using an asymmetric degree of freedom separation technique, where one person controls the rotation and the other one the translation of an object, as suggested by Pinho et al. [22] and used in several studies [1, 15, 22]. To achieve this, we designed the manipulation of an object to be exclusively accessible to one of the users. As it can be difficult for the AR user to manipulate the object's depth, since the tablet does not support stereoscopic images [22], we decided to always let the VR user translate the object. A single HTC Vice controller allows them to manipulate the object's position, by grabbing it and holding down the trigger button. In this isomorphic technique, the object now follows the movement of the user. During the translation process, the current orientation of the *movable object* is preserved to not enable the user to apply a rotation on their own. To prevent both users from manipulating simultaneously and therefore losing attention to each other's instruction, we implemented a locking mechanism. Due to this mechanism, the user, who is starting the manipulation first, captures the *movable object*. As proposed by Pinho et al., it is important to inform the other user about an object's state [22, 23], which is why the locked object is visualized by a transparent sphere with a lock symbol on it surrounding it on the other user's screen (See Figure 2 (b)).

In order to enhance communication between VR and AR users, we provide an alternative (or addition, depending on the condition) to natural speech as a method to transfer information. This alternative, which we will refer to as *widgets*, consists of visualizations of rotation and scaling requests. These widgets can be individually calibrated to the needed rotations and scale by the VR user and are then sent to the AR user. These widgets are displayed as a preview

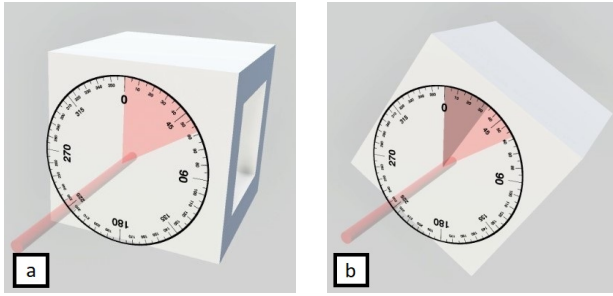


Figure 3: The AR user's view of the widgets before rotating the movable object (a) and after rotating it (b).

of the desired transformation, as well as a circle for rotation and a bar for scaling (See Figure 3). As soon as the widget configuration satisfies the user's desired manipulation, the widget can be sent to the AR user. For them, the widget is placed on top of the *movable object*. If the object is then manipulated by the AR user, the widget is filled with color for a better estimation of the desired rotation or scale (See Figure 3). Manipulation methods consist of rotation around the three axes and scaling. The AR user can switch between the different rotation axes with buttons and use a small touchpad area in the corner of the screen to manipulate the object with vertical and horizontal finger movements (See Figure 4).

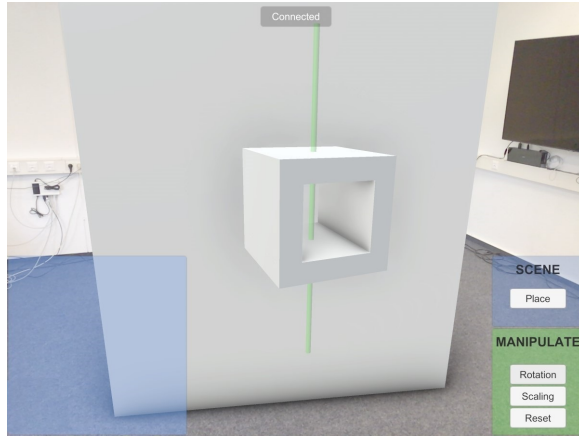


Figure 4: The view for the AR user.

2.2 Procedure

24 participants took part in the experiment, out of which 20 were male. They were aged between 21 and 59 years old ($M=34.33$, $SD=13.27$). Ten of them were students of IT-related fields. From the 12 participant pairs, ten knew each other before the study, while two pairs had little to no knowledge of each other. The students received course credit for their participation. Participants started by filling out a consent form. One of them was assigned to the iPad and the other one to the HMD. They did not switch roles, so the kind of technology was a between-subject factor whose evaluation is not part of the scope of this paper. The two users were located

in the same physical room, and the AR user was able to observe the VR user. Observation of the AR user by the VR user, on the other hand, was not possible; the only tracked position of the AR user was the tablet, and we therefore refrained from representing them in the VR scene altogether. First, the AR user had to align their view with that of the VR user. This was done manually, by aligning the scene with a fixed point on the floor of the physical room, and touching the “place”-button on the tablet (See Figure 4). According to their condition, they received a tutorial on how to control the input and how to act in the scene. When they felt comfortable, they started doing the experiment task with a training wall, followed by the first condition. The order of the conditions was counterbalanced between pairs using a Latin Square design to avoid an order bias. During the condition, video recordings were made and trial time, collisions, and usage of widgets were logged by the program automatically. After the participants finished, they filled out the System Usability Scale (SUS) [3] and had time to take a short break. Afterward, they continued with their second and third condition accordingly. When they had finished the third condition, they answered some questions about their experience as well as demographic questions.

3 RESULTS

Our results were evaluated using R. First, Shapiro-Wilk tests were conducted to test for normal distribution for all data. Since our data was not normally distributed, we then used Friedman and Wilcoxon signed-rank tests with a significance level of 0.05. The p-value of the pairwise comparisons was corrected using the Bonferroni method.

3.1 System Usability

The SUS was filled out after each condition. *Speech-Only* resulted in a mean score of 82.71 ($SD=11.07$), *Widgets-Only* received a mean score of 74.06 ($SD=15.25$). *Speech-and-Widgets* received the highest mean score with a value of 86.04 ($SD=13.08$). Testing the three conditions against each other with a Friedman test resulted in a p-value of 0.003. A Post-hoc test revealed a significantly higher mean score in *Speech-Only* in comparison to *Widgets-Only* ($p=0.03$). Also, the mean score of *Speech-and-Widgets* is significantly higher than the mean score of *Widgets-Only* ($p=0.003$). There was no significant increase between the mean scores of *Speech-Only* and *Speech-and-Widgets* ($p=0.27$). We can conclude that the participants preferred being able to use natural communication as well, instead of using the widgets only.

3.2 Personal Preferences

CONDITION	FASTEST	MOST PRECISE	MOST JOYFUL	WOULD USE AGAIN
<i>Speech-Only</i>	5	3	12	4
<i>Widgets-Only</i>	5	11	5	4
<i>Speech-and-Widgets</i>	14	10	8	16

Table 1: The number of people who perceived a certain condition as the fastest, the most precise one, the most joyful one and the one they would choose, if they had to do the task again.

In the post questionnaire, the participants were asked about which condition they preferred regarding speed, precision, and joy. In a fourth question we asked, which condition the participants would choose, if they had to do the experiment task again. The results are presented in Table 1. Most of the participants perceived the combination of speech and widgets as the condition they were the fastest with (58.3%). The condition voted for as most precise was *Widgets-Only* (45.8%), followed by *Speech-and-Widgets* (41.7%). 50% of the participants enjoyed using *Speech-Only* the most. The majority of votes regarding which condition to choose to solve the same task again was *Speech-and-Widgets* (66.7%).

3.3 Communication

To analyze the communication, we captured videos of the participants during the experiment. Two of the authors of this paper then manually watched the videos and inductively defined communication categories [2]. The inductive codes emerged from the data and were then used to count their occurrences in the different conditions. If the authors encountered disagreements in their coding, they discussed and rewatched the scene to come to an agreement. The final codes are:

- *Initiative* (INI): The AR user actively contributes to the solution without explicit requests (including hints and suggestions).
- *Miscommunication* (MISCOM): The VR user needs to correct/specify the request after the AR user reacted to it in an unexpected way.
- *Common ground mistake* (CGM): One of the users references scene objects that are not visible to the other user.
- *Exact requests* (EXACT): The VR user sends exact requests like “rotate for 20 degrees” or “rotate the object to form a diamond”. The usage of widgets is counted as an exact request as well.
- *Incremental requests* (INCR): The VR user requests to rotate for undefined amounts. Requests like “a bit more”, “further, further, stop” are counted as incremental requests.
- *Combination of exact and incremental requests* (COMBI): Exact requests followed by incremental requests for adjustment.

By using Wilcoxon tests, we found that the category *Incremental requests* occurred significantly more often in the *Speech-Only* condition. Exact requests, on the other hand, occurred more often in the *Speech-and-Widgets* condition. No significant differences could be found in the category *Combination of exact and incremental request*, *Initiative*, *Miscommunication*, and *Common ground mistake*. All the results can be seen in Figure 5 and Table 2.

Regarding the usage of widgets, the participants used more widgets in the *Widgets-Only* condition ($M = 25.17$, $SD = 8.86$) compared to the *Speech-and-Widgets* condition ($M = 13.17$, $SD = 6.13$). Post-hoc tests indicate that the scaling widget (*Widgets-Only*: $M = 12.58$, $SD = 4.44$, *Speech-and-Widgets*: $M = 4.5$, $SD = 4.36$, $p = 0.007$) and the widget displaying the y-rotation (*Widgets-Only*: $M = 5.33$, $SD = 2.46$, *Speech-and-Widgets*: $M = 3.17$, $SD = 1.64$, $p = 0.01$) are used significantly more often in the *Widgets-Only* condition than in the *Speech-and-Widgets* condition. The usage difference

COMMUNICATION CATEGORY	SPEECH-ONLY	SPEECH AND WIDGETS	P-VALUE
INI	1.5 ± 1.24	0.58 ± 1.17	0.053
MISCOM	2.83 ± 2.41	1.25 ± 1.14	0.057
CGM	1.17 ± 1.47	0.25 ± 0.45	0.11
EXACT	3.08 ± 2.64	10.67 ± 5.53	0.004
INCR	16.25 ± 7.71	6.25 ± 5.56	0.003
COMBI	2.92 ± 3.48	2.42 ± 3.12	0.72

Table 2: Mean and SD for the number of times that a certain communication pattern occurred in each category.

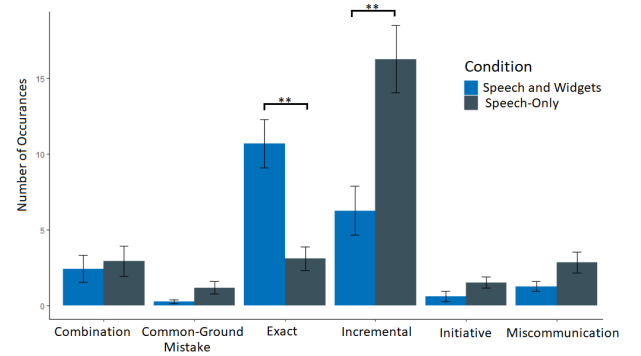


Figure 5: Occurrences of communication categories. Mean and standard error depicted. Significant results are marked with asterisks.

of x-rotation widgets (*Widgets-Only*: $M = 3.5$, $SD = 1.68$, *Speech-and-Widgets*: $M = 2.42$, $SD = 1.38$, $p = 0.09$) and z-rotation widgets (*Widgets-Only*: $M = 3.75$, $SD = 1.21$, *Speech-and-Widgets*: $M = 3.08$, $SD = 1.31$, $p = 0.4$) between the conditions was not significant. All results can be seen in Figure 6.

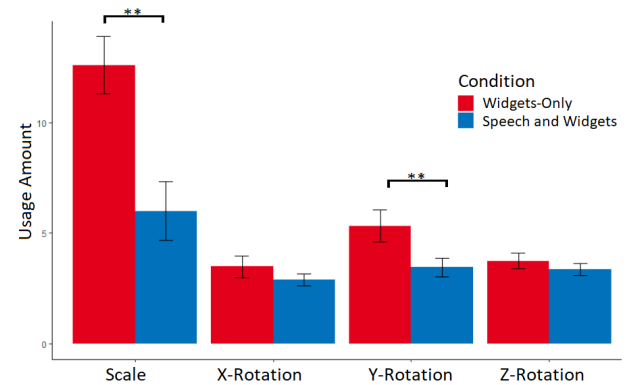


Figure 6: Usage of widgets for different purposes. Mean and standard error depicted. Significant results are marked with asterisks.

3.4 User Performance

During the study, we recorded the time needed to complete one trial and the number of collisions. In the *Speech-Only* condition, participants needed on average 68.51 seconds ($SD = 38.75$) to complete one trial and an average of 10 minutes and 16 seconds to complete the whole block. Participants needed the most time to complete one trial during the *Widgets-Only* condition, with a mean value of 111.76 ($SD = 97.09$) seconds per trial, and a total time of 16 minutes and 45 seconds. In the *Speech-and-Widgets* condition, the participants completed one trial in an average of 77.72 ($SD = 44.22$) seconds and the whole condition with a mean value of 11 minutes and 40 seconds. A significant difference between the three conditions could be found ($p < 0.001$). Post-hoc tests indicated that the trial time in *Widgets-Only* is significantly higher than in *Speech-Only* ($p < 0.001$) and in *Speech-and-Widgets* ($p = 0.002$). There was no significant difference between *Speech-Only* and *Speech-and-Widgets* conditions ($p = 0.6$). We can conclude that participants were generally faster when they were able to use speech as compared to only using widgets. We found a significant positive correlation between age and trial time ($p < 0.001$) and a significant negative correlation between hours played per week and trial time ($p < 0.001$). The gaming experience in general has significant effects on the trial time as well ($p < 0.001$).

On average, the participants had 6.8 collisions ($SD = 3.65$) during *Speech-Only* condition, 7.1 collisions ($SD = 5.28$) during *Widgets-Only* condition, and 5.4 collisions ($SD = 3.32$) during *Speech-and-Widgets* condition. A Friedman test presented no significant effects ($p = 0.54$).

4 DISCUSSION

The results of the study highlight the importance of speech as a human method of communication. While other channels like widgets can add to the amount of simultaneously transmitted information, speech is ubiquitous and serves as a general purpose tool for collaboration. During the experiment, both conditions allowing speech were rated significantly more usable than *Widgets-Only*. This is not surprising, as speech is a skill that participants were well accustomed to, rather than a new form of interaction with the introduced widgets. While speech provides the ability for fast correction of errors, sending widgets was received as cumbersome, resulting in a lower usability score. Usage of simpler widgets might enable quicker incremental adjustments, but with a trade-off in accuracy. When choosing the balance of speed and precision, the field of application needs to be considered. If a manipulation has to be carried out precisely, like in medicine or in assembly tasks [9, 19], the provided tools need to reflect the necessary level of accuracy. For all other tasks, we expect simple and easy to learn widgets to be the superior design.

Regarding the subjective perception of the participants, *Speech-Only* was perceived as the most pleasant experience. We suppose this might be because it directly increases the involvement between the users. The visual representation of the widgets, on the other hand, resulted in an increased perception of precision. Statistically, there were no significant differences in collisions between the conditions. This result contradicts our hypothesis H2 that widget usage results in higher precision. It did, however, confirm that visual representations gave the users confidence in their communications.

When both speech and widgets were utilized, participants could combine the measurable and perceived benefits of both interaction methods. The *Speech-and-Widgets* condition was perceived as the fastest interaction method. Even though the mean trial time for *Speech-Only* was lower than for *Speech-and-Widgets*, there was no significant difference between them. Both conditions involving speech were significantly faster than *Widget-Only*, confirming our hypothesis H1. The majority of participants stated that they would choose *Speech-and-Widgets* to complete the task again. The widgets were therefore perceived as a positive addition to natural speech, that require an unpleasant learning effort to be used. Their usage did not measurably improve precision or trial time, but had a positive effect on the users' confidence.

While widgets did not decrease the number of collisions, they changed the distribution of communication categories. Their usage reduced miscommunication and common ground mistakes, even though not significantly. Further, they replaced great portions of incremental requests by exact ones. This explains the increase in perceived precision. Widget usage did not improve the performance during interaction, but reshaped the communication process. Most miscommunications were about wrongly executing a request because there was no consensus about which axis to manipulate. Visualization of widgets reduced miscommunications, as consensus could be achieved more easily. Searching for landmarks or salient regions to describe a request, which were only visible to one of the technologies along the Reality-Virtuality continuum, led to referencing parts of the environment that were inconsistent with the other user's view. By using widgets, references could be directly created by the users. The usage of the widgets for scaling and y-rotation differed significantly between the *Widgets-Only* and *Speech-and-Widgets* conditions. The scaling widget could easily be replaced by speech, since there were no axes to be distinguished from each other. For rotations around the vertical axis, a strategy emerged which involved referencing the orientation of the wall by using speech. Finding consensus about the x-axis or z-axis proved to be a difficult task that can be achieved more easily by indicating the desired axis with widgets.

5 CONCLUSION

We can conclude that speech plays a fundamental role in communication between users collaborating in VR/AR interactions. In this scenario, the usage of widgets should only be seen as an extension rather than a replacement of language. Our first hypothesis, stating that conditions including speech would be completed more quickly, could be confirmed. In addition, the SUS scores for speech conditions underline the importance of speech in this collaborative manipulation task by showing greater usability, although we cannot confirm hypothesis H3. We had to reject our second hypothesis H2 regarding the increased precision of conditions amplified by widgets because there were no significant differences in collisions. It could be argued that collision is a poor measurement for precision. A distance measure between the *movable object* and hole center could have been a better way to predict precision. Although we were unable to confirm this hypothesis based on objective data, conditions with widgets received the majority of votes to be subjectively more precise.

The combination of speech and widgets causes the interaction between the users to be clearer and produces less miscommunication, as expected in our hypothesis H4, even though our low amount of data could not statistically confirm this. All in all, collaborative manipulation tasks between augmented and virtual reality should always include the possibility of speech as a communication tool. Natural language could be extended by visual representations to improve interaction. We believe that this conclusion could also be carried over to other collaborations on the Reality-Virtuality continuum, e.g., for immersive/non-immersive collaborations. For setups where both users are embodied, e.g., VR/VR, or VR/HMD-AR, gestures might make a difference in the communication style. These technological differences, as well as different tasks, or different asynchronous user roles, are worth exploring in future research. Finally, further research will improve our asynchronous/sequential interaction between users by leveraging concurrent and coincident interactions [20], i.e., simultaneously accommodating and translating a VR object while using speech to adjust it further.

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